Assignment 10:

Write a C program that defines a structure Rectangle with attributes length and width. Include a user-defined function named calculateArea with the signature float calculateArea(struct Rectangle r);. The function should calculate and return the area of the rectangle.

Source Code:

```
#include<stdio.h>
struct Rectangle {
   float length,width;
};
float calculateArea(struct Rectangle r) {
   return r.length * r.width;
}
int main() {
   struct Rectangle r;
   float area;
   printf("Enter the Length of the Rectangle : ");
   scanf("%f",&r.length);
   printf("\nEnter the width of the Rectangle : ");
   scanf("%f",&r.width);
   area = calculateArea(r);
   printf("The area of the Rectangle is %.2f",area);
   return 0;
}
```

Output:

```
Enter the Length of the Rectangle :5


Enter the width of the Rectangle :4


The area of the Rectangle is 20.00
Process finished with exit code 0
```

```
Enter the Length of the Rectangle :20


Enter the width of the Rectangle :15


The area of the Rectangle is 300.00
Process finished with exit code 0
```